SmartHull: Fixed-Stop Optical Fiber Sensing of Plate Bending

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Experiment Dates

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This experimental work was conducted at Applied Technology Schools – Abu Dhabi – Boys. The tests were carried out by the students (Saeed Hamad Sultan Hamad Alketbi, Rashed Rahmah Rashed Rahmah Alhantoobi and Hazza Ali Jafar Albani Alblooshi) under guidance and supervision of instructors: Nidal Saleh and Hamza Ayyash.

Abstract

Strain in a ship hull is normally measured with fiber Bragg grating (FBG) sensors, which need an interrogator that costs thousands of dollars. We tested whether a plastic optical fiber (POF) and an ESP32 can give repeatable estimates of plate bending strain at a small fraction of that cost. A POF was routed in two passes across a 297 × 210 × 2 mm polycarbonate plate. Two Fiber Opt Click modules read a sensing channel and a reference channel, and their ratio was used to cancel part of the drift. The plate was pressed to fixed stops at 8.5, 18.5 and 28.5 mm (nine presses). The ratio decrease rose from 0.93 % to 3.06 % across the stops. A straight-line fit gave $R^2 = 0.96$ and a slope of 0.111 ± 0.009 pp/mm ($p = 4 \times 10^{-6}$), and repeat presses agreed to 0.10 pp. A linear calibration estimated strain to ± 380 με (leave-one-out). A lack-of-fit test showed that the true curve is not exactly straight ($F = 34.5$, $p = 0.001$): the response rises faster between 8.5 and 18.5 mm than between 18.5 and 28.5 mm. The ratio drifted less than the raw sensing channel (0.055 against 0.093 %/min). The sensing hardware cost about US$80 at list prices, against US$700 to US$30,000 for FBG interrogators, although the two systems are not equivalent. The results are preliminary: nine presses, a plate stand-in and calculated, not measured, strain.

Introduction

Hulls flex under wave and cargo loads, and the strain this produces shows when a structure is being stressed beyond its design. FBG sensors are the standard way to measure strain in composite structures: they are accurate, can be multiplexed along one fiber and are immune to electrical noise [1]. Their cost is mostly the interrogator that reads the reflected wavelength, so a small school, workshop or boat builder rarely has one. POF sensors work on a different principle. Light from an LED travels along a large-core plastic fiber, and bending the fiber makes some of that light escape, so a simple photodiode can read the change as a voltage [2].

Our earlier report [3] showed that a POF routed across a plate gave a smaller receiver voltage as the plate was bent further, but the trials were hand-pressed, only five were recorded, and a single channel could not separate bending from drift of the LED and receiver. Intensity-based POF sensing has been proposed for structural health monitoring before [4, 5, 6]. What is still unproven for our arrangement is whether the response repeats well enough, and has the right shape, to be turned into a strain estimate. This report reports a repeatable test of that: fixed stops, a reference channel, nine presses and a statistical check of both the trend and the shape of the curve.

Research question and hypotheses

Research question. Can a low-cost intensity-based plastic optical fiber sensor, read by an ESP32, give repeatable estimates of plate bending strain between about 1,900 and 6,100 με?

H1 (trend). The sensing/reference ratio decrease q increases with centre deflection. The null hypothesis is a slope of zero. This is tested with a t test on the slope.
H2 (repeatability). Repeat presses at the same stop agree closely compared with the step between stops.
H3 (shape). A straight line is an adequate model of q against deflection, so that one calibration factor can turn q into strain. This is tested with a lack-of-fit F test, which compares the departure of the stop means from the line with the scatter of repeats. Bend-loss theory [7, 8] predicts a curved, not a straight, dependence on bend radius, so H3 is a real test and could fail.

H1, H2 and H3 were formed from the earlier trials and from theory; they are tested here on the nine fixed-stop presses. The null and the test for each are stated before the results are discussed, and every result is reported whichever way it falls.

Objectives

- To measure the sensing/reference ratio of a plastic optical fiber (POF) routed across a 297 × 210 × 2 mm polycarbonate plate at fixed-stop centre deflections of 8.5, 18.5 and 28.5 mm.
- To describe the relation between plate deflection and ratio decrease with a linear fit, to test whether the line has the right shape (lack of fit), and to assess repeatability and return to baseline.
- To convert the ratio decrease into strain with a linear calibration and report its leave-one-out error.
- To compare the drift of the sensing/reference ratio with the drift of the raw sensing channel.
- To relate published hull strain limits to the plate by beam equations and set fixed alert levels.
- To compare the cost of the sensing hardware with that of an FBG system.

Theory

The plate rests on two supports and is pressed at its centre to a fixed stop. The centre deflection, δ, is the downward movement of the centre from its unloaded position. Each stop was set to a fixed height, so δ is known for every press.

Optical response quantities

The ratio of the sensing channel (pin D35) to the reference channel (pin D34) has no units:

$$R = \frac{V_{\text{sensing, D35}}}{V_{\text{reference, D34}}} \tag{1}$$

With R_0 the unloaded ratio and R_b the ratio while bent, the ratio decrease is:

$$q\ (\%) = 100\left(1 - \frac{R_b}{R_0}\right) \tag{2}$$

A positive value means less light reached the sensing channel.

Why bending reduces the light (macrobending loss)

A step-index POF guides light by total internal reflection at the boundary between the core (refractive index n_1) and the cladding ($n_2 < n_1$). A ray stays guided if its angle of incidence at the boundary, measured from the normal, is larger than the critical angle, $\sin \theta_c = n_2/n_1$. For typical PMMA POF, $n_1 \approx 1.49$ and $n_2 \approx 1.40$, so $\theta_c \approx 70°$ [3]. These are typical catalogue values, not measured for our fiber. In a straight fiber, every guided ray meets the wall at more than θ_c (Figure 1a).

In a bend the geometry changes. A ray that reaches the outer wall meets it at a smaller angle of incidence than it had on the straight, because the wall now turns away from the ray. For a ray that enters a bend of centreline radius R at radius r_0, travelling at angle α to the axis, the angle of incidence θ_i at the outer wall (radius R + a, with a the core radius) obeys $\sin \theta_i = (r_0/(R + a)) \cos \alpha$. If θ_i falls below θ_c, the ray refracts into the cladding and is lost instead of being reflected [6, 7]. Steeper rays (larger α) and tighter bends (smaller R) lose more, and the loss is not a straight-line function of curvature in theory [7, 8]. When the receiver sees less light, the ratio R_b/R_0 falls and q rises.

On the plate, the fiber follows the surface, so the bend radius is set by the deflection. With the centre curvature $\kappa = 2\varepsilon/t$ from the strain model, the centre radius is about 515, 242 and 163 mm at 8.5, 18.5 and 28.5 mm (Figure 1b). These radii are large compared with the 1 mm fiber, so the loss is small, of the order of 1 to 3 %, which is what we measure. Other effects may add to it, such as strain in the fiber and tape and changes in launch conditions; this experiment cannot separate them from bend loss.

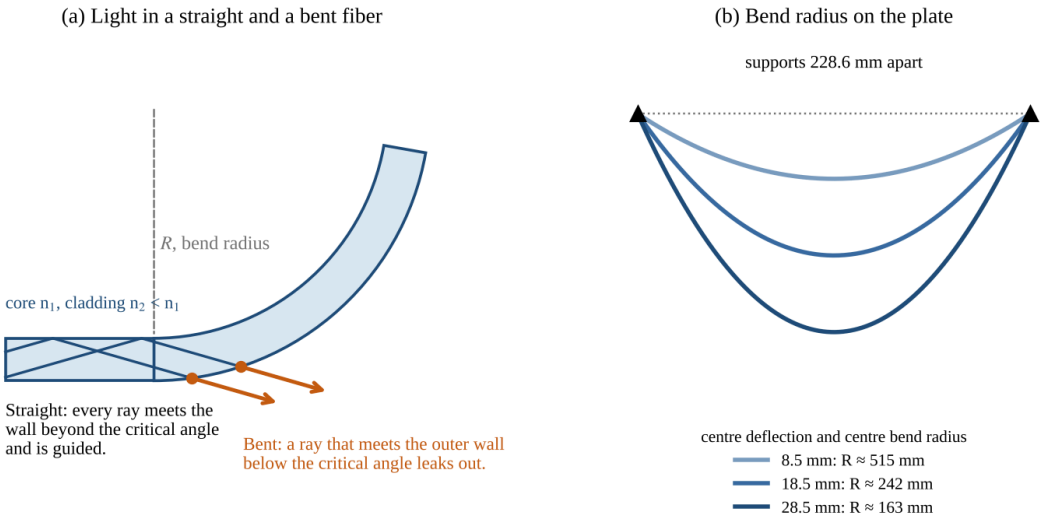


Figure 1. (a) Light in a straight and a bent step-index fiber: the guided ray (blue) is reflected at every wall hit, while rays that meet the outer wall of the bend below the critical angle leak out (orange). The drawing is a schematic with the critical angle of a typical PMMA fiber (70°). (b) Plate profile at the three fixed stops (schematic) with the centre bend radius from the strain model, R = t/(2ε).

Bending strain

For small deflection, the centre surface strain of a beam of thickness t on supports L apart, loaded at the centre, is:

$$\varepsilon = \frac{6\,t\,\delta}{L^2} \tag{3}$$

Here t = 2 mm and L = 228.6 mm, giving 229.6 με per mm. Large deflections are treated with a finite-rotation (elastica) beam model [9, 10], where the surface strain is half the thickness times the curvature:

$$\kappa(x) = \frac{|w''(x)|}{\left[1 + (w'(x))^2\right]^{3/2}}, \qquad |\varepsilon_b| = \frac{t}{2}\,\kappa \tag{4}$$

This is a geometric result and needs no elastic modulus. It models an ideal beam, not the measured fiber strain.

Linear fit and error measures

The decrease is fitted to deflection by least squares:

$$\hat{q} = a\,\delta + b \tag{5}$$

With $\bar{q}$ the mean observed decrease and n the number of presses, the coefficient of determination and the root-mean-square error are:

$$R^2 = 1 - \frac{\sum_i (q_i - \hat{q}_i)^2}{\sum_i (q_i - \bar{q})^2} \tag{6}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (q_i - \hat{q}_i)^2} \tag{7}$$

RMSE is in percentage points (pp). Both are in-sample measures, not held-out accuracy.

Strain calibration

The decrease is turned into strain with a single linear factor c fitted to the elastica strains:

$$\varepsilon_{\mathrm{est}} = c\, q \tag{8}$$

Its error is estimated by leave-one-out: each press is left out, c is refitted, and the left-out strain is predicted.

Alert levels

The deflection that corresponds to a limit strain follows from equation (3), and the stress in the plate from Hooke's law with E = 2,300 MPa:

$$\delta = \frac{\varepsilon L^2}{6\, t} \tag{9}$$

$$\sigma = E\, \varepsilon \tag{10}$$

Drift

Drift is the slope of a signal over a quiet 60-second window, in % per minute, taken for the sensing channel, the reference channel and the ratio.

Lack of fit

R^2 says how much of the spread a line explains, but not whether the line is the right shape. When there are repeat presses at each of k stops, the residual sum of squares of the line can be split into pure error, SS_{PE}, which is the scatter of repeats about their own stop mean (n − k degrees of freedom), and lack of fit, SS_{LOF}, which is how far the stop means lie from the line (k − 2 degrees of freedom). Their ratio is compared with the F distribution [11]:

$$F = \frac{SS_{\mathrm{LOF}}/(k-2)}{SS_{\mathrm{PE}}/(n-k)} \tag{11}$$

A small p-value means the stop means depart from the line by more than the repeat scatter can explain, so the line is not the right shape. The test assumes independent presses with roughly normal errors and equal spread at each stop.

Design iterations

The final arrangement is the result of five changes, each made to fix a problem seen in the previous one. The values below are from the earlier report [3] and from this report.

1. **Bare fiber bends.** The fiber was bent by hand with no plate. Receiver voltage fell 8.5 to 10.7 % on steep bends, so the fiber responds, but one release came back 4.8 % above

baseline after the fiber shape changed, and the bend radius was not measured. *Change:* use a defined geometry.

2. **Fiber on a plate, one channel.** The fiber was taped across a polycarbonate plate and the plate was pressed by hand to 10, 20 and 30 mm read with a ruler. The voltage fell 2.06 % at 20 mm and 3.63 % at 30 mm, but the 10 mm trial rose 1.27 % and a changing bend gave 7.73 %. Loads of 900 g gave only 0.54 to 0.58 % increases and were dropped. *Problem:* one voltage cannot separate bending from drift of the LED and receiver. *Change:* add a reference.

3. **Reference channel and ratio.** A second Fiber Opt Click module with a separate fiber loop was added, and the ratio $R = V_{sensing}/V_{reference}$ was used. Five trials gave decreases of 1.19 to 4.00 %, but released readings differed from baseline by −0.86 to +1.35 %, and hand-set deflections were not repeatable. *Change:* fix the deflection.

4. **Fixed stops.** The plate was pressed to stops at 8.5, 18.5 and 28.5 mm in a dark room. Compared with free-hand bends, R^2 rose from 0.764 to 0.96 and RMSE fell from 0.476 to 0.18 pp; repeat presses agreed to 0.10 pp. *New problem:* only two presses at two stops, and the shape of the curve is not yet known (see Lack of fit).

5. **Live page, calibration and alert levels.** The ESP32 streams about 10 rows per second to a live page that follows the baseline slowly, converts q to strain with $\varepsilon = 1{,}917.3\ q$ and shows fixed amber and red levels. The code, data and page are public.

Apparatus

- ESP32 microcontroller
- Two MikroElektronika Fiber Opt Click modules (sensing and reference channels)
- Plastic optical fiber (sensing path and separate reference loop)
- Polycarbonate plate, 297 × 210 × 2 mm
- Two supports, spacing ≈ 9 in (228.6 mm)
- Fixed stops for the plate centre at 8.5, 18.5 and 28.5 mm
- Laptop running the SmartHull live page, which logs the readings as CSV

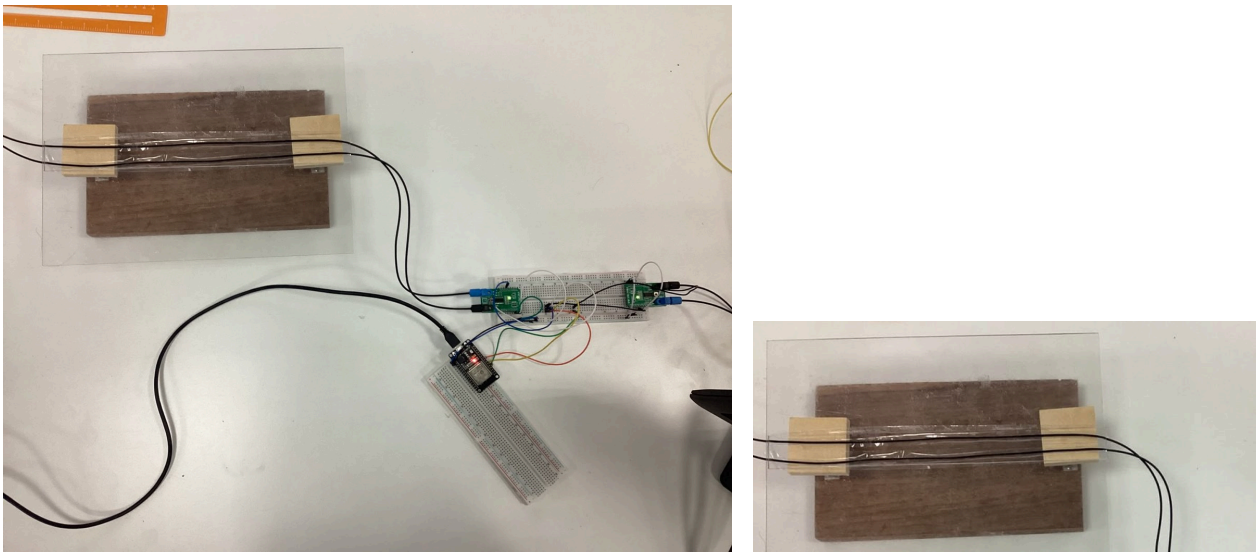


Figure 2. Test rig. (a) The polycarbonate plate on two supports, with the sensing fiber across it, and the ESP32 and two Fiber Opt Click modules on breadboards. (b) Top view of the plate with the fiber taped across it, one pass across and one return pass.

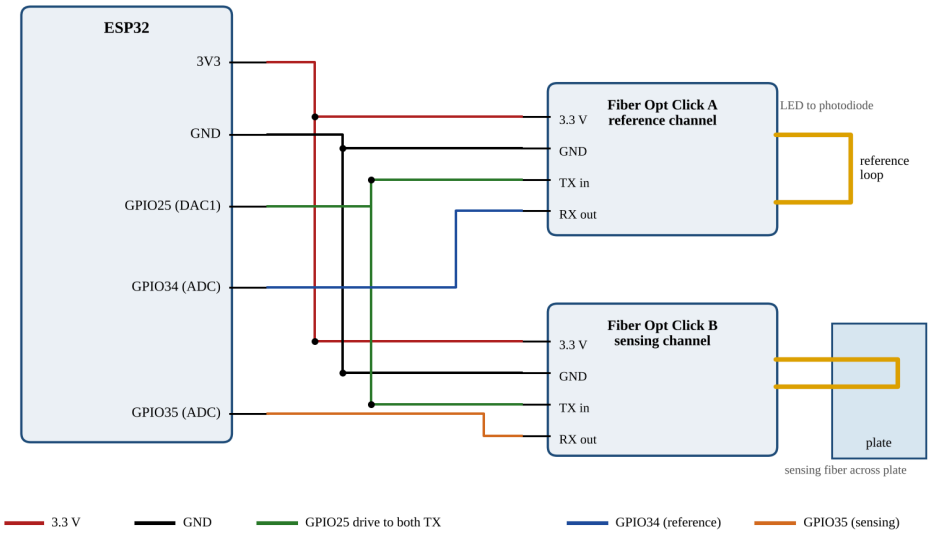


Figure 3. Circuit diagram. Both modules are powered from 3.3 V with a common ground and both TX inputs are driven by ESP32 pin D25 (DAC1). The reference module (A) reads on D34 and the sensing module (B) on D35; each module has its own LED and photodiode, linked by the fiber loop.

Procedure

1. Power both Fiber Opt Click modules from 3.3 V with a common ground, and drive both TX inputs from ESP32 pin D25 at command setting 132/255.
2. Connect the reference loop to D34 and the sensing path to D35.
3. Route the sensing fiber across the plate: one pass across, a turn at one end and a return pass, forming one continuous sensing path. Place the plate on the two supports.
4. Record an unloaded baseline R_0 before each press.

5. Press the plate centre down to a fixed stop (8.5, 18.5 or 28.5 mm), hold it, and record the ratio R_b.
6. Release the plate and record the released ratio to check the return to baseline.
7. Repeat to obtain nine presses: five at 8.5 mm, two at 18.5 mm and two at 28.5 mm. (Table 1)
8. Calculate R and q using equations (1) and (2), fit q against δ with equation (5), and calculate R^2 and RMSE using equations (6) and (7). (Table 2) Split the residual into pure error and lack of fit and calculate F with equation (11). (Table 5)
9. Calculate the strain at each stop with the elastica model, equation (4), and fit the calibration factor of equation (8) with leave-one-out error.
10. Record 43 quiet 60-second windows with no bending, and calculate the drift of the sensing channel, the reference channel and the ratio. (Table 4)

Data

Fixed-stop plate trials

Table 1. Fixed-stop presses. Decrease q from equation (2); strain from the elastica model, equation (4).

Deflection δ (mm)	Presses (n)	Ratio decrease q per press (%)	Mean q (%)	Calculated strain (με)
8.5	5	1.00, 0.78, 0.87, 0.99, 1.0	0.93	≈ 1,940
18.5	2	2.45, 2.37	2.41	≈ 4,132
28.5	2	3.02, 3.09	3.06	≈ 6,135

The calculated strain is an ideal-beam estimate at the set deflection; it is not a measured plate or fiber strain. The nine presses are separate loading cycles; repeated rows within one hold are not independent trials.

Fit, calibration and repeatability

Table 2. Fit and error measures for the nine fixed-stop presses.

Quantity	Value
Slope a, equation (5)	0.111 ± 0.009 pp/mm (standard error)
R^2, equation (6)	0.96 (r = 0.98)
Slope t statistic, p-value	13.0, $p = 4 \times 10^{-6}$
Slope 95 % confidence interval	0.091 to 0.131 pp/mm
RMSE, equation (7)	0.18 pp
Calibration factor c, equation (8)	1,917.3 με per 1 % decrease
Leave-one-out strain error	± 380 με
Calibrated range	0–6,500 με (lowest tested point 1,940 με)
Repeat standard deviation	0.10 pp
Return to baseline after release	within 0.07 %

Comparison with free-hand bends

Table 3. Fixed stops against free-hand bends.

Method	R^2	RMSE (percentage points)
Fixed stops (9 presses)	0.96	0.18
Free-hand bends	0.764	0.476

Drift with no bending

Table 4. Median absolute drift over 43 quiet 60-second windows.

Signal	Median absolute drift (%/min)
Sensing channel	0.093
Reference channel	0.059
Ratio, equation (1)	0.055

The ratio drifted less than the raw sensing channel in 77 % of the windows.

Lack of fit

Table 5. Split of the residual of the straight line into pure error and lack of fit (equation (11)), and the strain error each stop mean would give with the single calibration factor.

Source	Sum of squares (pp²)	Degrees of freedom	Mean square	F	p-value
Straight line (regression)	7.393	1	7.393		
Lack of fit (stop means vs line)	0.2595	1	0.2595	34.5	0.0011
Pure error (repeats within stops)	0.0451	6	0.00752		
Total residual	0.3046	7			

Stop (mm)	Presses	Mean q (%)	Line (%)	Mean − line (pp)	SE of mean (pp)	Strain from mean q (με)	Calculated strain (με)	Error
8.5	5	0.93	0.99	−0.06	0.039	1,779	1,940	−161 (−8 %)
18.5	2	2.41	2.10	+0.31	0.061	4,621	4,132	+489 (+12 %)
28.5	2	3.06	3.21	−0.15	0.061	5,857	6,135	−278 (−5 %)

The pure-error standard deviation is 0.087 pp. The calibration factor c = 1,917.3 με per 1 % is applied to each stop mean; the calculated strain is from the elastica model.

Results

The ratio decrease rises with deflection across the three fixed stops and is close to a straight line, although a lack-of-fit test (below) shows it is not exactly straight. The all-press linear fit (n = 9, descriptive $R^2 = 0.96$) and the calibration are:

$$q\ (\%) \approx 0.111\,\delta\ (\text{mm}) + 0.046, \qquad \varepsilon_{\text{est}}\ (\mu\varepsilon) \approx 1917.3\,q \tag{12}$$

Statistical support for the trend

Four measures from the fit agree that the decrease is a real, increasing function of deflection, well summarised by a straight line, and not noise:

$R^2 = 0.96$	$r = 0.98$	$t = 13.0$	0.18 pp
96 % of the variation in q is explained by deflection	correlation coefficient	slope / standard error, p = 4 × 10^{-6}	RMSE, about 6 % of the 3.06 % largest mean decrease

$$t_{\text{stat}} = \frac{a}{SE(a)} = \frac{0.111}{0.0085} = 13.0 \quad (\nu = n - 2 = 7) \tag{13}$$

The slope is 0.111 pp/mm with a standard error of 0.0085 pp/mm, so its 95 % confidence interval is 0.091 to 0.131 pp/mm (t distribution, 7 degrees of freedom). The interval is far from zero: the slope is about 13 standard errors above it, and the probability of seeing such a trend by chance, if there were no relation, is about 4 in a million. The repeat standard deviation of 0.10 pp is smaller than the 0.93 pp step between the first two stops, so adjacent stops are clearly separated. These

results support H1 and H2. They do not test the shape of the curve, which is the subject of the next section.

What the statistics mean

In plain terms, pressing the plate further reliably makes the sensor read a larger drop in light, and the size of that drop grows in proportion to the deflection: about 0.11 percentage points for every extra millimetre. $R^2 = 0.96$ means that deflection alone accounts for 96 % of the spread in our readings, leaving only 4 % to noise and everything else, and $r = 0.98$ says the points lie close to a straight line (although the next section shows the true curve bends slightly). The very small p-value says the trend is not a coincidence of nine lucky points, and the narrow confidence interval says the slope is well pinned down, between 0.091 and 0.131 pp/mm. The RMSE of 0.18 pp is the typical gap between a reading and the line, and it is small next to the signal, so a reading can be turned back into a deflection and a strain. What the statistics do not mean is that the sensor is already accurate on a real hull: they describe this plate, these nine presses and the range 8.5–28.5 mm, and the ± 380 με leave-one-out error is the honest size of the strain uncertainty.

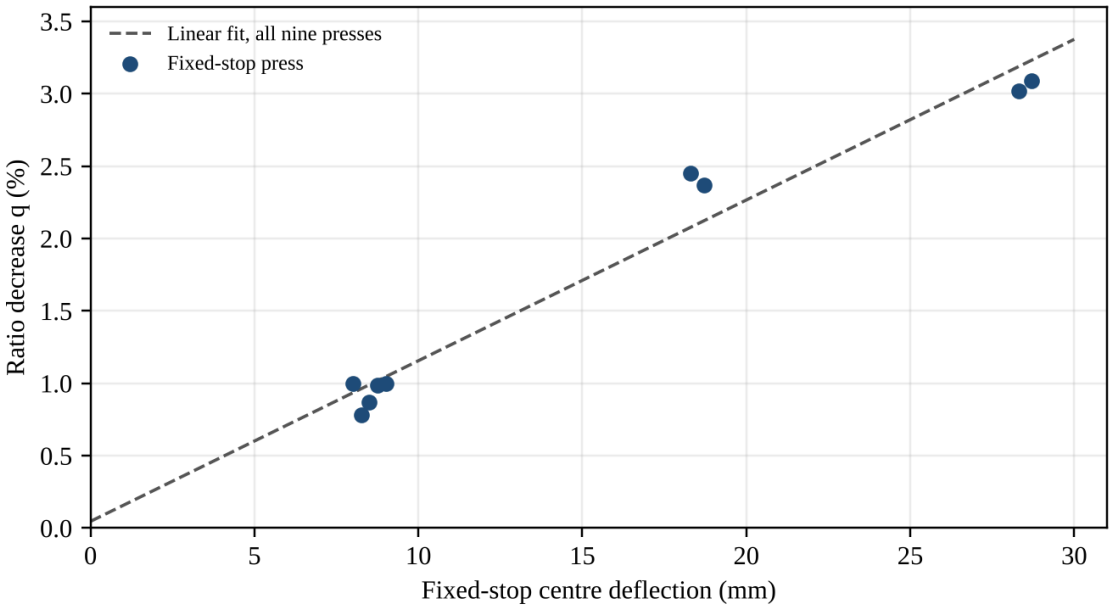


Figure 4. Ratio decrease versus fixed-stop centre deflection. All nine presses are shown, with small horizontal offsets so repeats do not overlap. The dashed line is the least-squares fit, drawn over 0–30 mm.

A 1 % decrease is about 1,917 με, with a leave-one-out error of ± 380 με. Repeat presses agree to 0.10 pp and the ratio returns to within 0.07 % of baseline. Free-hand bends fit much worse ($R^2 = 0.764$, RMSE 0.476 pp), so the fixed stops matter. These are preliminary descriptions over the sampled range, and R^2 describes agreement within these presses, not held-out prediction accuracy.

The curve is not exactly straight (lack of fit)

A high R^2 does not show that a line is the right shape, so H3 was tested directly. Table 5 splits the residual of the line (0.3046 pp², 7 degrees of freedom) into pure error from the repeats (0.0451 pp², 6 degrees of freedom) and lack of fit (0.2595 pp², 1 degree of freedom). The result is $F(1, 6) = 34.5$, $p = 0.001$: **H3 is rejected**. The stop means lie further from the line than the repeat scatter (0.087 pp) can explain.

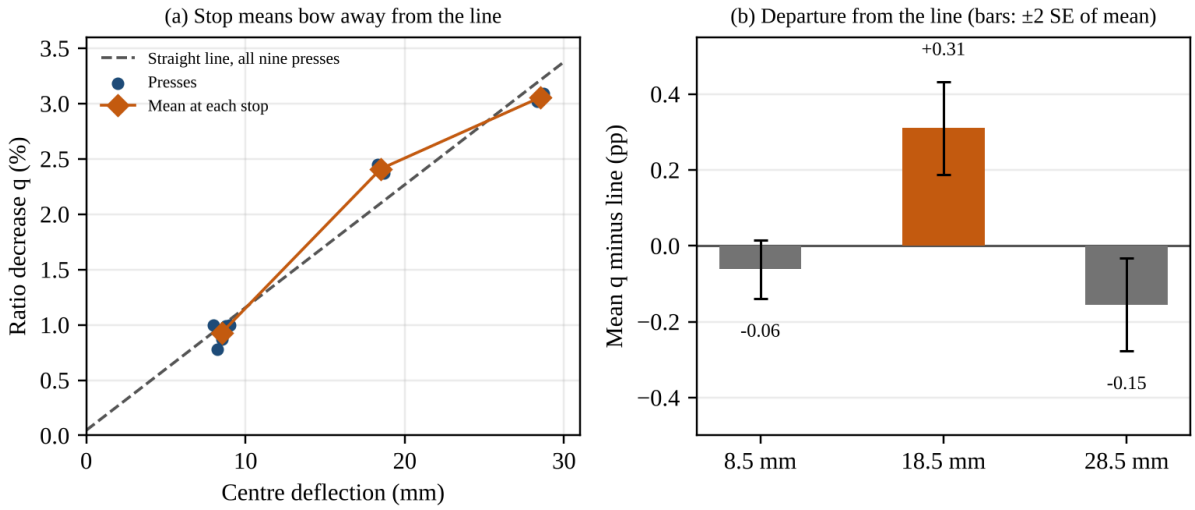


Figure 5. (a) The nine presses, the straight-line fit and the mean at each stop. (b) Departure of each stop mean from the line, with ± 2 standard errors of the mean from the pure-error spread.

The finding is about the shape of the curve. The mean at 18.5 mm is 0.31 pp above the line, about five standard errors, while the means at 8.5 and 28.5 mm lie 0.06 pp and 0.15 pp below it. The slope between stops is 0.148 pp/mm from 8.5 to 18.5 mm but only 0.065 pp/mm from 18.5 to 28.5 mm, so the response flattens at larger deflections. The strain model cannot explain this: the strain per millimetre of deflection falls by only about 6 % over the same range (228 to 215 με/mm). The flattening is therefore in the optical response. Counting the unloaded baseline (q = 0 at δ = 0, which is by definition and not a press), the slope from 0 to 8.5 mm is 0.109 pp/mm, which would make the curve S-shaped, but with one point below 8.5 mm this is a suggestion and not a test.

What it means for strain. The trend remains statistically supported (H1) and the line remains a good summary ($R^2 = 0.96$). But the single calibration factor misreads the stops by −161, +489 and −278 με (−8, +12 and −5 %), so the worst case, 489 με, is about a quarter of the 2,000 με red level. The ± 380 με leave-one-out error is typical of this scatter and is consistent with it, but it should not be read as a bound at every stop. A curve with three parameters would pass exactly through three stop means, so it cannot be validated with these data; more stops are needed.

Possible causes (not tested). (i) Bend loss in the regime of these large radii is not linear in curvature in theory [7, 8], and the steepest rays, which leak first, may be used up so that less remains to lose at larger deflection. (ii) The tape and fiber may slip or stiffen at larger deflection. (iii) The contact of the stop and the plate width, which the beam model ignores. This experiment cannot tell these apart.

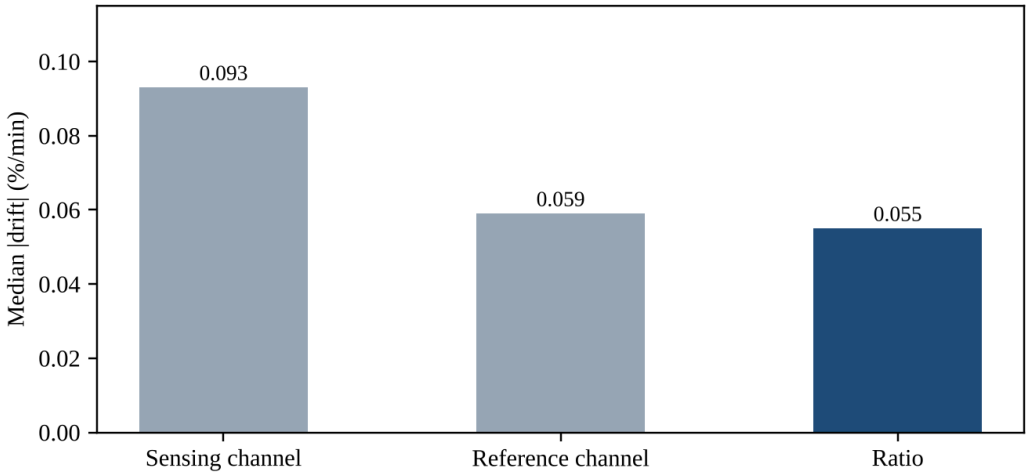


Figure 6. Median absolute drift over 43 quiet 60-second windows. The ratio drifts less than the raw sensing channel and less than the reference channel.

The ratio cancels part of the common drift, but 0.055 %/min is not zero. Over several minutes it can add up to a visible fraction of the 1 % signal at the lowest stop, which is why the live page follows the baseline slowly.

Strain limits and alert levels

The alert levels are fixed from published limits, not set by the user. Glass-reinforced plastic (GRP) starts micro-cracking at about 0.2 % strain [12]. BS 4994:1987 allows the lesser of 0.1 ε_R and 0.2 % [13], but it is a tank and vessel standard, so it is used only as a cross-check. The red level is therefore 2,000 με, and amber is 80 % of it, 1,600 με (our own choice). Equation (9) gives the plate deflection that matches each level, and equation (10) the plate stress.

Table 6. Alert levels with equivalent plate deflection and stress (t = 2 mm, L = 228.6 mm, E = 2,300 MPa).

Level	Strain (με)	Plate deflection (mm)	Plate stress (MPa)
Amber (APPROACHING)	1,600	7.0	3.7
Red (DANGER)	2,000	8.7	4.6

At red, the plate stress is about 3 % of the 60 MPa yield, so the plate is never at risk. It is only a curvature stand-in; the limit belongs to a GRP hull. The small-deflection slope of 229.6 με/mm and the elastica values of 228, 222 and 214 με/mm at 10, 20 and 30 mm sit close to the calibrated 213 με/mm (0.111 × 1,917).

Statistical integrity

The fit statistics are strong, and with n = 9 they show a clear, significant increase that a straight line summarises well ($R^2 = 0.96$, $p = 4 \times 10^{-6}$), while the lack-of-fit test shows that the true curve bends ($p = 0.001$). They are reported with their limits, so that the claim made is exactly what the data support. The rules we followed, and what each result does and does not support, are listed below.

- **All data shown.** All nine fixed-stop presses appear in Table 1 and Figure 4. No press was removed after seeing the results, and no exclusion rule was applied, unlike the filtered subset in the earlier report, which was chosen after inspecting the data.
- **Small sample.** n = 9 presses at three heights (5, 2 and 2). The 18.5 mm and 28.5 mm stops have only two presses each, so their means are weakly determined and no confidence interval is claimed for them.
- **Presses, not rows.** Each press is one trial. Repeated samples within a hold are not independent and were not counted as extra trials.
- **In-sample fit.** R^2 = 0.96 and RMSE = 0.18 pp describe agreement within these nine presses. They are not held-out prediction accuracy. R^2 is high partly because the three stops are far apart.
- **Lack of fit.** The F test uses a pure-error spread from 6 degrees of freedom, 4 of them from the five presses at 8.5 mm, and assumes roughly normal errors with equal spread at each stop. With three stops, it can say the line is the wrong shape but not what the right shape is.
- **Out-of-sample estimate.** The ± 380 με figure is leave-one-out, which is a more honest estimate of prediction error than the in-sample RMSE, but it is still from the same nine presses.
- **Units.** RMSE and repeat standard deviation are in percentage points of the ratio decrease. They are not relative errors of 0.18 % or 0.10 %.
- **Range.** The calibration is valid only inside the tested range. The lowest tested point is 1,940 με, so the amber level (1,600 με) is an extrapolation.
- **Strain labels are modelled.** The calibration is fitted to elastica strains, which are calculated, not measured. The calibration error therefore does not include any error in the beam model.
- **Drift windows.** The 43 quiet windows come from one recording session, so they are not fully independent. Medians are reported because they are less affected by outlying windows, and no uncertainty is claimed for them.
- **Where significance is claimed.** The only test used is the t test on the slope of equation (5), which assumes roughly normal, independent residuals. The 77 % drift figure is the share of windows in which the ratio drifted less than the sensing channel, not a tested difference.

Result	Supported	Not supported
Decrease rises with deflection	Clear trend over 8.5–28.5 mm	Behaviour below 8.5 mm or above 28.5 mm
Linear fit, $R^2 = 0.96$, $p = 4 \times 10^{-6}$	A statistically significant increasing trend within these nine presses, well summarised by a line	Held-out prediction accuracy
Lack of fit, $F = 34.5$, $p = 0.001$	The stop means depart from the straight line; the response flattens at larger deflections	The exact curve shape, or its cause, with only three stops
Strain, ± 380 με	Leave-one-out error on the calibrated range	Measured strain, or any hull strain
Ratio drifts less than raw channel	Lower median drift, and in 77 % of windows	A tested, significant difference

Cost comparison with FBG systems

Cost was part of the motivation, so we compared the list price of the hardware in one sensing channel with the price of an FBG system. Prices are retail or marketplace listings found on 6 November 2026, not vendor quotes, and the plate and supports are excluded because they stand in for the hull.

Table 7. Hardware cost of one sensing channel, SmartHull against FBG (US dollars, list prices).

Item	SmartHull (this work)	FBG system
Sensor	POF, assumed 3 m at US$0.75 per metre ≈ US$2 [15]	Packaged FBG strain sensor US$50–200 each; bare FBG US$10–50 [17]
Light source and receiver	Two Fiber Opt Click boards at US$34.95 = US$69.90 [14]	Included in the interrogator
Readout and logging	ESP32 board ≈ US$10 (3,569 HUF, excluding VAT) [16]	Interrogator: US$700–30,000; complete multichannel listings US$2,300–19,000 [17]; “well into thousands” [18]
Total, one sensing channel	**≈ US$80**	**≈ US$750 to US$30,000**

On these prices the SmartHull hardware is about 30 times cheaper than the cheapest complete interrogator listing we found (US$2,300) and more than 200 times cheaper than a US$19,000 system. The comparison is not like for like. An FBG measures strain directly, with a wavelength signal that does not depend on source power, and several gratings share one fiber and one interrogator, so the cost per point of an FBG system falls as points are added, while each extra SmartHull sensing point needs another receiver module. Our system must be calibrated against calculated strain, is accurate to about ± 380 με on a plate, and drifts. The two are therefore different tools: the intensity sensor is for low-cost screening and teaching, and does not replace an FBG where strain accuracy matters. The Click boards are a development choice, and a custom board would probably cost less, but we have not priced one.

Conclusion

Results

Fixed-stop pressing at 8.5, 18.5 and 28.5 mm gave mean ratio decreases of 0.93, 2.41 and 3.06 %, a linear fit with $R^2 = 0.96$ ($r = 0.98$), RMSE 0.18 pp and slope 0.111 ± 0.009 pp/mm ($t = 13.0$, $p = 4 \times 10^{-6}$), which statistically supports an increasing and close-to-linear relation between deflection and ratio decrease. A lack-of-fit test ($F = 34.5$, $p = 0.001$) shows that the curve is not exactly straight: it rises faster between 8.5 and 18.5 mm than between 18.5 and 28.5 mm. Repeat presses agreed to 0.10 pp and the ratio returned to within 0.07 % of baseline. A linear calibration, $\varepsilon = 1{,}917.3\ q$, estimates strain to ± 380 με (leave-one-out) over 0–6,500 με. The ratio drifted 0.055 %/min against 0.093 %/min for the raw sensing channel, and was lower in 77 % of quiet windows. Alert levels of 1,600 and 2,000 με come from the published GRP micro-cracking limit and correspond to 7.0 and 8.7 mm on the plate.

Limitations and sources of error

- The plate is polycarbonate and stands in for hull curvature; the strain limit is for GRP.
- Only nine presses at three heights; the lowest tested point is 1,940 με, so the 1,600 με amber level lies below the tested range.
- Calculated strain is an ideal-beam estimate, not a measured fiber strain. The plate width-to-span ratio is about 0.92 and the stop contact is not a point, so these effects are not captured by the beam model.
- The two sensing passes form one optical path; pass spacing and turn radius were not measured.
- Movement of the fiber, supports or connectors, and unequal drift of the two emitters and receivers can shift readings.
- Fit and leave-one-out errors are in-sample, small-sample measures, not independent validation.
- Results are small-sample and in-sample (see Statistical integrity); the calibration is linear although the response is slightly curved; no AI or machine-learning model is used in this work.

Recommendations and further improvements

- Add fixed stops at 2–7 mm and at least 10 presses at five or more heights, to cover the amber level and to map the curve, then fit and validate a curved calibration on held-out presses.
- Check the strain labels with an independent strain gauge or profile measurement, and test a real GRP coupon.
- Record longer tests of sensing-only against sensing-to-reference readings under controlled source changes.
- Use the linear calibration as the baseline and compare polynomial regression, random forest and a compact multilayer perceptron on whole-cycle training, validation and test splits, judged by held-out MAE and RMSE and by ESP32 memory and speed.

- Deploy a validated lightweight model on the ESP32 within the tested range; extension to hull materials, several sensing locations and marine conditions is a later objective.

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